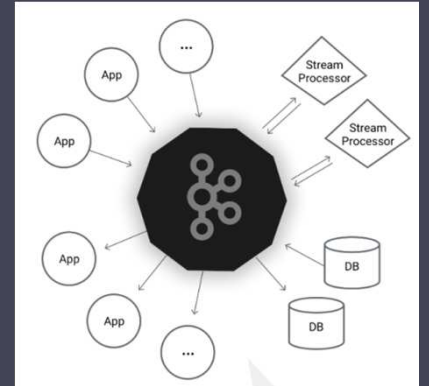


Apache Kafka

Introduction



Objectives

- History
- Introduction

History

- Started as an internal infrastructure system built at LinkedIn
- LinkedIn has lots of databases and other systems built to store data, an architecture was something that would help us to handle the continuous flow of data was missing.
- Project started from scratch
 - Instead of holding piles of data like our relational databases, key-value stores, search indexes, or caches
 - Focus on treating data as a continually evolving and ever growing stream, and build a data system – and indeed a data architecture
- The project aims to provide a unified, high-throughput, low-latency platform for handling real-time data feeds.
- The development team at LinkedIn was led by Jay Kreps
- Kafka was released as an open source project on GitHub in late 2010

Introduction

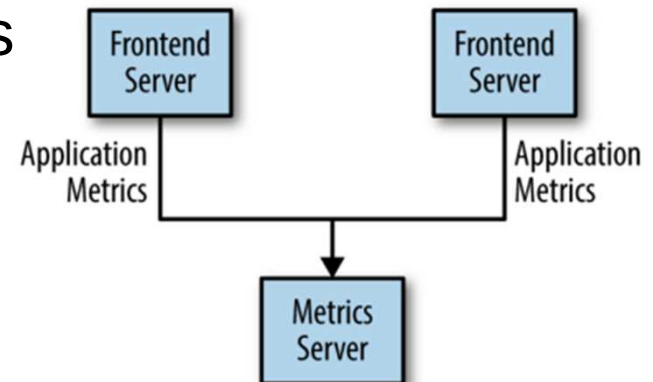
- Every application creates data (log messages, metrics, user activity, outgoing messages and more)
 - Every byte of data has a story to tell
 - Something of importance informing the next thing to be done
- Kafka is a streaming platform
 - A system that lets you publish and subscribe to streams of data, store them, and process them
- Kafka is like a messaging system, but not yet another queue like other messaging systems.
 - Works as a modern distributed system, running on a cluster capable of scaling all applications even in massive companies.
 - Kafka is a true storage system built to store data for as long as you might like.
 - World of stream processing raises the level of abstraction quite significantly.

Publish / Subscribe

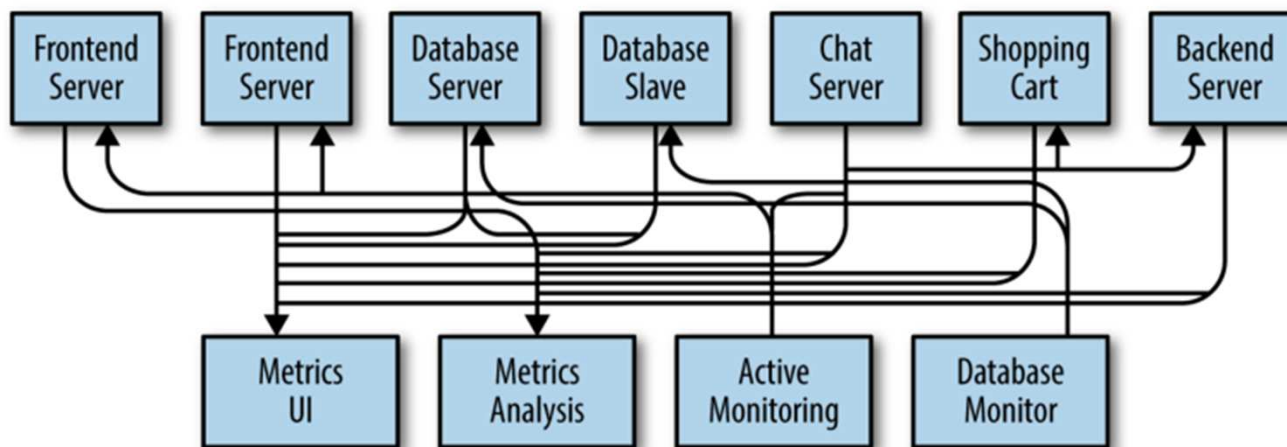
- Messaging is a pattern that is characterized by the sender (publisher) of a piece of data (message) not specifically directing it to a receiver.
- The publisher classifies the message somehow, and that receiver (subscriber) subscribes to receive certain classes of messages.
- Pub/sub systems often have a broker, a central point where messages are published, to facilitate this

How it started

- Consider an application that needs to send monitoring information metrics using direct connection on dashboard.

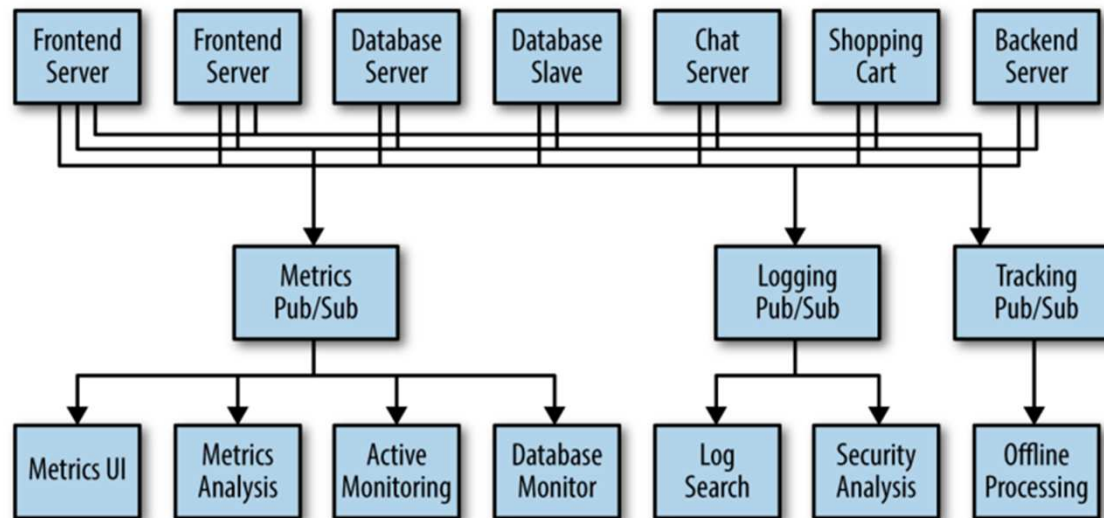


- As time proceeds we have more application added changing the architecture like this, it is harder to trace.



How it started...

- Individual Queue Systems – By introducing this architecture now looks more simple
- But limitation is, there is a lot of duplication.

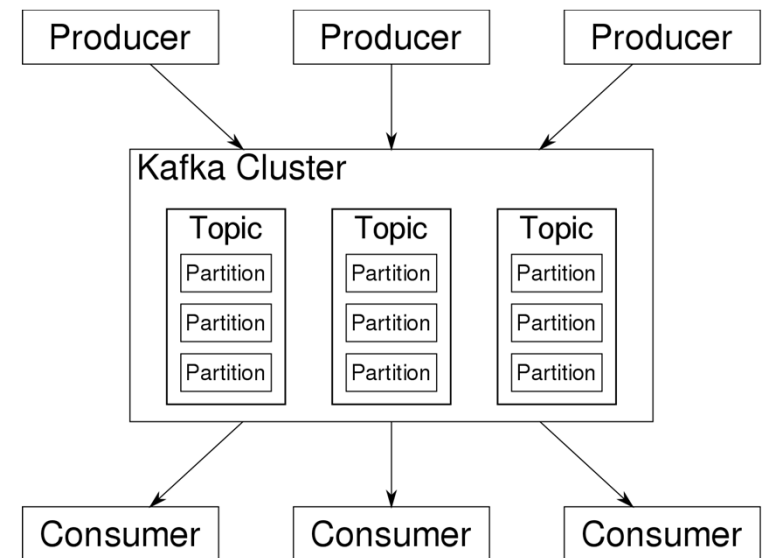


Publish/Subscribe Messaging

- Publish Subscribe is a messaging pattern where senders of messages, called publishers, do not program the messages to be sent directly to specific receivers, called subscribers, but instead categorize published messages into classes without knowledge of which subscribers.
- Similarly, subscribers express interest in one or more classes and only receive messages that are of interest, without knowledge of which publishers.
- Pub/sub systems often have a broker, a central point where messages are published

Apache Kafka

- Apache Kafka is a publish/subscribe messaging system designed to solve the problem of individual queue systems.
- Kafka is often referred as 'distributing streaming platform'
- Kafka stores the data durably, in order and can be read deterministically.
- A key feature of Apache Kafka is that of retention, which is the durable storage of messages for some period of time.



Summary

- History
- Introduction